POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5024A | Semester V | Theory | 4 Credits

Non-destructive Testing

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5024A as Non-destructive Testing in Semester V for Mechanical Engineering. The official SITTTR detailed source was unavailable. With user approval, this handbook uses reliable public NDT curriculum sources and does not represent recovered official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. Training or this handbook alone does not constitute personnel certification/qualification; regulated NDT, radiation work and interpretation require approved procedures, supervision and applicable standards.

Verified source note: Sources cover visual, penetrant, magnetic-particle, radiographic and ultrasonic methods, calibration, interpretation and limits of education versus certification.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Non-destructive Testing evaluates materials, components and assemblies for discontinuities without destroying serviceability. It combines method selection, procedure control, equipment checks, interpretation, reporting and strict safety discipline.

Malayalam support: □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□;
□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□
□□□□□□□□.

Prerequisite foundation

Engineering materials, workshop practice, basic waves/electricity and safety awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain NDT purpose, limitations, standards and personnel roles.
2. Explain visual, liquid-penetrant and magnetic-particle methods.
3. Explain radiographic and ultrasonic-testing principles and limitations.
4. Select a conceptual method and communicate findings responsibly under approved procedures.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain NDT purpose, discontinuities, method selection and qualification limits.	Understand
CO2	Explain VT, PT and MT principles, applications and limitations.	Understand
CO3	Explain RT and UT principles, calibration and interpretation concepts.	Understand
CO4	Prepare a conceptual inspection plan/report using approved procedures and safety controls.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	NDT Foundations and Visual Testing	Purpose, discontinuities, method selection, standards, qualification and visual inspection.	Approx. 14 hours	CO1	Module 1
2	Liquid Penetrant and Magnetic Particle	Surface-breaking indication methods, process steps, sensitivity, material limits and interpretation.	Approx. 16 hours	CO2	Module 2
3	Radiographic and Ultrasonic Testing	Radiation imaging, UT wave propagation, calibration, indications, sensitivity and safety.	Approx. 16 hours	CO3	Module 3
4	Inspection Planning, Reporting and Ethics	Procedure selection, calibration, traceability, acceptance criteria, reporting, quality systems and safety.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

NDT Foundations and Visual Testing

Purpose, discontinuities, method selection, standards, qualification and visual inspection.

CO1 · Approx. 14 hours

Liquid Penetrant and Magnetic Particle

Surface-breaking indication methods, process steps, sensitivity, material limits and interpretation.

CO2 · Approx. 16 hours

Radiographic and Ultrasonic Testing

Radiation imaging, UT wave propagation, calibration, indications, sensitivity and safety.

CO3 · Approx. 16 hours

Inspection Planning, Reporting and Ethics

Procedure selection, calibration, traceability, acceptance criteria, reporting, quality systems and safety.

CO4 · Approx. 14 hours

MODULE 1

NDT Foundations and Visual Testing

Purpose, discontinuities, method selection, standards, qualification and visual inspection.

Approx. 14 hours | CO1 | Understand · Apply

Learning goals

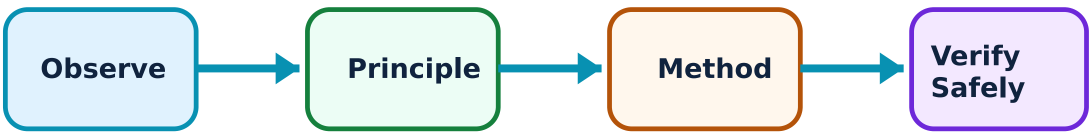
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for NDT Foundations and Visual Testing: identify the system, state its principle, follow the correct method and verify the result safely.

1. NDT purpose and limits−

NDT detects or characterises selected discontinuities/properties while preserving the item. Method capability depends on material, geometry, access, defect orientation and procedure.

Study and application method

State component, material, expected flaw type/location, access and acceptance basis before selecting a method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State component, material, expected flaw type/location, access and acceptance basis before selecting a method.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: An NDT result is valid only within procedure, calibration, coverage and personnel-qualification limits.

2. Visual testing−

VT uses direct/remote visual examination, lighting, gauges and surface condition to identify accessible indications and dimensional/weld concerns.

Study and application method

Define viewing/lighting/access conditions, inspect systematically and record location, scale, orientation and evidence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define viewing/lighting/access conditions, inspect systematically and record location, scale, orientation and evidence.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Common mistake / precaution: Cleanliness and adequate lighting are essential; absence of a visual indication does not prove absence of internal defects.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Liquid Penetrant and Magnetic Particle

Surface-breaking indication methods, process steps, sensitivity, material limits and interpretation.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

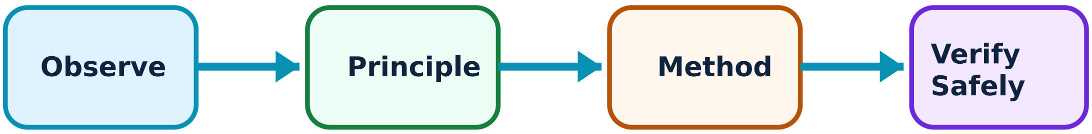
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Liquid Penetrant and Magnetic Particle: identify the system, state its principle, follow the correct method and verify the result safely.

1. Liquid penetrant testing−

PT uses capillary action to reveal surface-breaking discontinuities in suitable nonporous materials after cleaning, penetrant, dwell, removal and developer steps.

Study and application method

Follow approved procedure sequence and record material, surface condition, dwell, lighting and indication evaluation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Follow approved procedure sequence and record material, surface condition, dwell, lighting and indication evaluation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: PT does not reveal subsurface flaws and chemicals require SDS/PPE/ventilation controls.

2. Magnetic particle testing−

MT magnetises ferromagnetic materials; leakage fields near surface/near-surface discontinuities attract particles to form indications.

Study and application method

Identify material suitability, magnetisation direction, particle method and demagnetisation/cleaning requirement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify material suitability, magnetisation direction, particle method and demagnetisation/cleaning requirement.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: MT is not suitable for non-ferromagnetic materials; field direction affects detectability.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Radiographic and Ultrasonic Testing

Radiation imaging, UT wave propagation, calibration, indications, sensitivity and safety.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

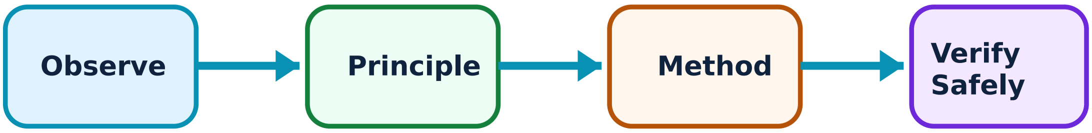
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Radiographic and Ultrasonic Testing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Radiographic testing−

RT uses X- or gamma-radiation and film/digital imaging to show density/thickness variations and internal discontinuities under controlled geometry.

Study and application method

At a conceptual level identify source, object, detector, exposure geometry, image-quality control and interpretation record.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: At a conceptual level identify source, object, detector, exposure geometry, image-quality control and interpretation record.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Radiation work is tightly regulated: only authorised trained personnel under approved radiation-safety controls may perform it.

2. Ultrasonic testing−

UT sends high-frequency sound into a component and interprets reflected/transmitted signals to locate features or measure thickness. Coupling, angle, calibration and geometry affect results.

Study and application method

State probe/mode, reference standard, calibration, scan coverage and signal-evaluation basis under an approved procedure.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State probe/mode, reference standard, calibration, scan coverage and signal-evaluation basis under an approved procedure.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: UT indications require trained interpretation; geometry and orientation can create false/missed indications.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Inspection Planning, Reporting and Ethics

Procedure selection, calibration, traceability, acceptance criteria, reporting, quality systems and safety.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

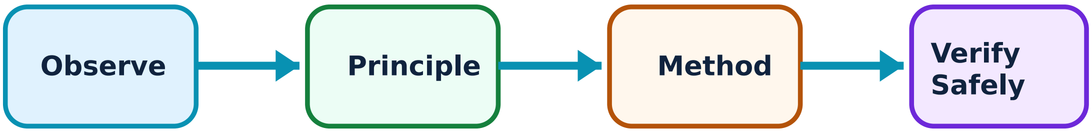
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Inspection Planning, Reporting and Ethics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Inspection plan−

An inspection plan defines item identification, method/procedure, equipment/calibration, coverage, acceptance criteria, personnel and records.

Study and application method

Create a conceptual plan with scope, method rationale, safety controls, reference standard and result disposition path.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a conceptual plan with scope, method rationale, safety controls, reference standard and result disposition path.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Never invent acceptance criteria; use the authorised specification/code.

2. Reporting and integrity−

Reports must be traceable, factual and clear about coverage, settings, indications, limitations and reviewer. Data integrity protects public and industrial safety.

Study and application method

Record item/procedure/revision, equipment/calibration, examiner, conditions, results, limitations and approval status.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Record item/procedure/revision, equipment/calibration, examiner, conditions, results, limitations and approval status.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Common mistake / precaution: Do not alter results or imply certification beyond actual authorisation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: NDT Foundations and Visual Testing

Use the concept flow to revise observation, principle, method and verification.

Module 2: Liquid Penetrant and Magnetic Particle

Use the concept flow to revise observation, principle, method and verification.

Module 3: Radiographic and Ultrasonic Testing

Use the concept flow to revise observation, principle, method and verification.

Module 4: Inspection Planning, Reporting and Ethics

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare method suitability for surface versus internal flaws in stated materials.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draft a conceptual VT/PT inspection record for an instructor-provided sample.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Interpret a supplied UT/RT schematic only under instructor guidance.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a safety and qualification checklist for a simulated NDT assignment.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — NDT Foundations and Visual Testing

Explain NDT purpose and limits and state one correct method, application or precaution.−

NDT detects or characterises selected discontinuities/properties while preserving the item. Method capability depends on material, geometry, access, defect orientation and procedure.

Answer framework: State component, material, expected flaw type/location, access and acceptance basis before selecting a method.

Important point: An NDT result is valid only within procedure, calibration, coverage and personnel-qualification limits.

Explain Visual testing and state one correct method, application or precaution.−

VT uses direct/remote visual examination, lighting, gauges and surface condition to identify accessible indications and dimensional/weld concerns.

Answer framework: Define viewing/lighting/access conditions, inspect systematically and record location, scale, orientation and evidence.

Important point: Cleanliness and adequate lighting are essential; absence of a visual indication does not prove absence of internal defects.

Module 2 — Liquid Penetrant and Magnetic Particle

Explain Liquid penetrant testing and state one correct method, application or precaution.−

PT uses capillary action to reveal surface-breaking discontinuities in suitable nonporous materials after cleaning, penetrant, dwell, removal and developer steps.

Answer framework: Follow approved procedure sequence and record material, surface condition, dwell, lighting and indication evaluation.

Important point: PT does not reveal subsurface flaws and chemicals require SDS/PPE/ventilation controls.

Explain Magnetic particle testing and state one correct method, application or precaution.−

MT magnetises ferromagnetic materials; leakage fields near surface/near-surface discontinuities attract particles to form indications.

Answer framework: Identify material suitability, magnetisation direction, particle method and demagnetisation/cleaning requirement.

Important point: MT is not suitable for non-ferromagnetic materials; field direction affects detectability.

Module 3 — Radiographic and Ultrasonic Testing

Explain Radiographic testing and state one correct method, application or precaution.

−

RT uses X- or gamma-radiation and film/digital imaging to show density/thickness variations and internal discontinuities under controlled geometry.

Answer framework: At a conceptual level identify source, object, detector, exposure geometry, image-quality control and interpretation record.

Important point: Radiation work is tightly regulated: only authorised trained personnel under approved radiation-safety controls may perform it.

Explain Ultrasonic testing and state one correct method, application or precaution.−

UT sends high-frequency sound into a component and interprets reflected/transmitted signals to locate features or measure thickness. Coupling, angle, calibration and geometry affect results.

Answer framework: State probe/mode, reference standard, calibration, scan coverage and signal-evaluation basis under an approved procedure.

Important point: UT indications require trained interpretation; geometry and orientation can create false/missed indications.

Module 4 — Inspection Planning, Reporting and Ethics

Explain Inspection plan and state one correct method, application or precaution.−

An inspection plan defines item identification, method/procedure, equipment/calibration, coverage, acceptance criteria, personnel and records.

Answer framework: Create a conceptual plan with scope, method rationale, safety controls, reference standard and result disposition path.

Important point: Never invent acceptance criteria; use the authorised specification/code.

Explain Reporting and integrity and state one correct method, application or precaution.−

Reports must be traceable, factual and clear about coverage, settings, indications, limitations and reviewer. Data integrity protects public and industrial safety.

Answer framework: Record item/procedure/revision, equipment/calibration, examiner, conditions, results, limitations and approval status.

Important point: Do not alter results or imply certification beyond actual authorisation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: NDT Foundations and Visual Testing.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Liquid Penetrant and Magnetic Particle.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Radiographic and Ultrasonic Testing.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Inspection Planning, Reporting and Ethics.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Purpose, discontinuities, method selection, standards, qualification and visual inspection..
2. Develop a complete answer or practical plan for: Surface-breaking indication methods, process steps, sensitivity, material limits and interpretation..
3. Develop a complete answer or practical plan for: Radiation imaging, UT wave propagation, calibration, indications, sensitivity and safety..
4. Develop a complete answer or practical plan for: Procedure selection, calibration, traceability, acceptance criteria, reporting, quality systems and safety..

LAST-MILE REVISION

Quick Revision

Module 1: NDT Foundations and Visual Testing

Purpose, discontinuities, method selection, standards, qualification and visual inspection.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Liquid Penetrant and Magnetic Particle

Surface-breaking indication methods, process steps, sensitivity, material limits and interpretation.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Radiographic and Ultrasonic Testing

Radiation imaging, UT wave propagation, calibration, indications, sensitivity and safety.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Inspection Planning, Reporting and Ethics

Procedure selection, calibration, traceability, acceptance criteria, reporting, quality systems and safety.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
NDT purpose and limits	NDT detects or characterises selected discontinuities/properties while preserving the item.
Visual testing	VT uses direct/remote visual examination, lighting, gauges and surface condition to identify accessible indications and dimensional/weld concerns.
Liquid penetrant testing	PT uses capillary action to reveal surface-breaking discontinuities in suitable nonporous materials after cleaning, penetrant, dwell, removal and developer steps.
Magnetic particle testing	MT magnetises ferromagnetic materials; leakage fields near surface/near-surface discontinuities attract particles to form indications.
Radiographic testing	RT uses X- or gamma-radiation and film/digital imaging to show density/thickness variations and internal discontinuities under controlled geometry.
Ultrasonic testing	UT sends high-frequency sound into a component and interprets reflected/transmitted signals to locate features or measure thickness.
Inspection plan	An inspection plan defines item identification, method/procedure, equipment/calibration, coverage, acceptance criteria, personnel and records.
Reporting and integrity	Reports must be traceable, factual and clear about coverage, settings, indications, limitations and reviewer.

LEARNING RESOURCES

References

Recommended NDT references

1. ASNT, Nondestructive Testing Handbook.
2. J. Prasad and C. G. K. Nair, Non-Destructive Testing and Evaluation of Materials.
3. Applicable approved NDT procedures and codes.

Approved public NDT sources

- https://catalog.tri-c.edu/course-descriptions/zndt/zndt.pdf
- https://www.monroeccc.edu/sites/default/files/programs/files/NDT/NonDestructiveTesting-2019-20n.pdf

These sources support the study handbook while official Course 5024A detail is unavailable; they do not certify personnel or replace approved procedures.